

Combinatorics

Permutations: $P(n, r) = \frac{n!}{(n-r)!}$

Permutations of k types of indistinguishable objects: $\frac{n!}{n_1! n_2! \dots n_k!}$

Combinations: $C(n, r) = \frac{n!}{(n-r)! r!}$

Probability

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Bayes' Theorem: $P(B|A) = \frac{P(B|A) \cdot P(A)}{P(B)}$

Discrete Random Variables:

CDF: $F(x) = \sum_x F(x)$

Expectation: $E(X) = \sum_x x f(x)$, $E(ax+b) = aE(X) + b$, $E(X) + E(Y) = E(X+Y)$

Variance: $V(X) = E(X^2) - E(X)^2$, $V(ax+b) = a^2 V(X)$

Law of the Unconscious Statistician: $E(h(x)) = \sum_x h(x) f(x)$

Continuous Random Variables

CDF: $P(X \leq x) = \int_{-\infty}^x f(t) dt$, $P(a \leq X \leq b) = \int_a^b f(t) dt$

Expectation: $E(X) = \int_{-\infty}^{+\infty} x f(x) dx$, $E(ax+b) = aE(X) + b$, $E(X) + E(Y) = E(X+Y)$

Variance: $V(X) = E(X^2) - E(X)^2$, $V(ax+b) = a^2 V(X)$

Law of the Unconscious Statistician: $E(h(x)) = \int_{-\infty}^{+\infty} h(x) f(x) dx$

Probability Distributions

Bernoulli: p = probability of successful trial

Binomial: n = number of trials
 p = probability of successful trial

Geometric: p = probability of successful trial
 X denotes number of trials until (and including) the first success.

Negative Binomial: p = probability of successful trial
 X denotes number of trials until (and including) the r^{th} success

Poisson: λ = expected number of occurrences in a given time period

Exponential: λ = expected number of occurrences in a given time period

Normal: μ = mean
 σ = standard deviation

Standard Normal: $Z = \frac{x - \mu}{\sigma}$, $X \sim N(\mu, \sigma) \Rightarrow Z \sim N(0, 1)$

Normal Approximation:
of Binomial: $\mu = np$
 $\sigma^2 = np(1-p) \Rightarrow \sigma = \sqrt{np(1-p)}$
— Continuity correction needed

Normal Approximation:
of Poisson: $\mu = \lambda$
 $\sigma^2 = \lambda$
— Continuity correction needed

Poisson Approximation:
of Binomial: $\lambda = np$

Joint Random Variables

CDF: Discrete: $F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$
Continuous: $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$
 $\Rightarrow P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$ } order of integrals does not matter.

Marginal: Discrete: $f_X(x) = P(X=x) = \sum_y f(x, y)$
Dist.

Continuous: $f_X(x) = P(X=x) = \int_{-\infty}^{+\infty} f(x,y) dy$

Conditional Dist: $f_{X|Y=y}(y) = P(X=x | Y=y) = \frac{f(x,y)}{f_Y(y)}$

Expectation: $\left. \begin{array}{l} \text{Discrete: } E(X) = \sum_x x f_X(x) \\ \text{Continuous: } E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx \end{array} \right\} E(aX+bY) = aE(X) + bE(Y)$

Variance: $V(X) = E(X^2) - E(X)^2$, $V(aX+bY+c) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cor}(X,Y)$

Law of the Unconscious Statistician: $E(h(X,Y)) = \sum_x \sum_y h(x,y) f(x,y)$

Covariance: $\text{Cov}(X,Y) = E(XY) - E(X)E(Y)$, $\text{Cov}(aX+b, cY+d) = ac \text{Cov}(X,Y)$

Correlation Coefficient: $\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sqrt{V(X)V(Y)}}$

Statistics

Markov's Inequality: $P(X \geq a) \leq \frac{E(X)}{a} \quad \forall a > 0$

Chebyshev's Inequality: $P(|X-\mu| \geq k) \leq \frac{\sigma^2}{k^2} \quad \forall k > 0$

Distribution of Sample mean $\bar{x}$: $E(\bar{x}) = \mu$, $V(\bar{x}) = \frac{\sigma^2}{n} \Rightarrow \text{s.d.}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

Central limit Theorem: As $n \rightarrow \infty$, $\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$

Confidence Intervals + Hypothesis Testing

Standard error: Sample mean: $\frac{\sigma}{\sqrt{n}}$
Sample proportion: $\sqrt{\frac{P(1-P)}{n}}$

Confidence interval: Two-sided: $\text{Statistic} \pm |z_{\alpha/2}| \times \text{Standard error}$
One-sided: $\text{Statistic} \pm |z_{\alpha}| \times \text{Standard error}$

Type I error: $\alpha = \text{probability of } H_0 \text{ being true when we reject it}$

Type II error: $\beta = \text{probability of } H_0 \text{ not being rejected when it is false}$

Power: $1-\beta$